

Jacobi-Digit Deformations of the Fabius–Rvachev Law

Geometric Bessel products, spherical random flights, Rayleigh cumulants,
higher-order refinement, and inverse endpoint asymptotics

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Abstract

This report develops a two-parameter deformation of the normalized Fabius–Rvachev probability law. The uniform “digits” in the usual geometric random series are replaced by symmetric beta, equivalently Jacobi, digits

$$\rho_\alpha(u) = \frac{\Gamma(\alpha + \frac{1}{2})}{\sqrt{\pi} \Gamma(\alpha)} (1 - u^2)^{\alpha-1} \mathbf{1}_{(-1,1)}(u), \quad \alpha > 0,$$

and one studies

$$X_{q,\alpha} = (1 - q) \sum_{n \geq 0} q^n U_{\alpha,n}, \quad 0 < q < 1.$$

The case $\alpha = 1$ is the geometric-uniform family already developed in the repository; $q = 1/2, \alpha = 1$ is the normalized Rvachev up law. The new parameter changes the one-step atom rather than merely the geometric scale. It turns the sinc product into an infinite product of normalized Bessel functions and produces a continuous bridge among Bernoulli convolutions, arcsine/Bessel cascades, the Rvachev law, spherical random flights, Jacobi–Gegenbauer polynomials, and Hermite asymptotics.

The principal results proved here are: an exact geometric Bessel-product formula and super-polynomial Fourier decay; a spherical random-flight model with an annulus-to-ball support transition; Rayleigh-sum cumulants and Bell moments with a clean q -integer factor; an exact Bessel spectral-zeta factorization; a countable dense set of scale resonances and a generic simple-zero theorem; a universal Weyl law for the Fourier zeros; finite higher-order refinement equations for every integer α ; an exact fractional Volterra–pantograph equation at an endpoint; a uniform Wasserstein-rate desingularization of Bernoulli convolutions; Gaussian limits along both $q \uparrow 1$ and $\alpha \rightarrow \infty$; and a Jacobi–Hermite orthogonal-polynomial bridge. The endpoint small-ball law implies a new inverse-quantile asymptotic, including the usual inverse Fabius leading scale as a specialization.

A symbolic/numerical study reveals an unexpected turning transition in the third monic recurrence coefficient at $\alpha_3^* \approx 0.066515108060995$. This phase statement, a complete Lambert- W_{-1} endpoint expansion, algebraic nonresonance of Bessel zeros, and a matrix-valued Thue–Morse generalization are isolated as conjectures or research programs. Novelty claims throughout are repository-relative, not claims of global historical priority.

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1 Scope, status, and repository audit

1.1 Snapshot and audit method

The final repository comparison in this report is pinned to commit

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of [ProveIt](#). At that snapshot, a GitHub code-index query returned 119 files with extension `.tex` under `Analysis/FabiusFunction/docs`. The corpus was audited by a combination of:

- (i) direct reading of the canonical consolidated volumes, their manifests, and the active frontier reports most closely related to the present construction;
- (ii) repository-wide exact searches for the defining phrases and formulas of each proposed result; and
- (iii) specialization checks against the established geometric-uniform, sinc-product, endpoint, inverse-Fabius, orthogonal-polynomial, and spectral-zeta theories.

The repository moved during the investigation, so a commit hash is essential for a reproducible “not already present” statement.

The nearest existing materials include the canonical exponents and q -series volume, the cyclotomic q -Fabius frontier, the parameter-response report, the generalized-base atomic-function report, the representation volume, and the inverse-endpoint reports. In particular, the repository already contains the uniform-digit family

$$(1 - q) \sum_{n \geq 0} q^n U_n, \quad U_n \sim \text{Unif}[-1, 1],$$

its sinc product, Bernoulli cumulants, Bell moments, Gaussian limit, Legendre-to-Hermite transition, and the dyadic Thue–Morse sign law. Those facts are used as consistency checks but are not presented as new results.

1.2 Novelty crosswalk

Nearest repository theme	Already present	Distinct contribution here
Geometric-uniform q -law	sinc product, cumulants, plateau/Gaussian regimes, q -Appell and spectral constructions	Jacobi digits; normalized Bessel factors; a second deformation parameter changing the atom law
General exponent sequences	arbitrary geometric/nongeometric scales applied to fixed atoms	fixed geometric scales applied to a beta/Jacobi continuum of atoms, including a sphere-dimension interpretation
Bessel representations	Bessel formulas used to represent or transform the fixed up law	a Bessel product that <i>defines</i> a new probability family, with Rayleigh cumulants and Bessel-zero resonances
Orthogonal polynomials	Legendre, native up-law, and Hermite limits for established laws	a two-parameter Gegenbauer/Jacobi-to-Hermite flow and a new small- α turning phase
Inverse endpoint asymptotics	high-order dyadic uniform/Fabius expansions using Lambert W_{-1}	a fractional-order endpoint equation and α -dependent inverse-quantile scale
Bernoulli convolutions	scale limits and related Bernoulli-convolution phenomena	a smooth Jacobi-digit regularization with an explicit W_1 convergence rate as $\alpha \downarrow 0$

Table 1: Repository-relative distinction of the present frontier.

Exact indexed searches for beta-digit, Jacobi-digit, Jacobi-Hermite bridge, and the defining random-flight phrase returned no matches in the pinned corpus. This is strong evidence against exact repository duplication, but it is not a literature-wide priority claim.

1.3 Status convention

Statements marked **proved here** are proved in the report, possibly using standard external theorems explicitly cited. Statements marked **numerical** are supported by the included reproducible program and data. Statements marked **conjectural** are intended as precise research targets. A specialization may be known even when the parameter-uniform statement is new; such specializations are identified rather than relabeled.

2 The Jacobi-digit cascade

Throughout the report,

$$0 < q < 1, \quad a := 1 - q, \quad \alpha > 0, \quad \nu := \alpha - \frac{1}{2}, \quad L := \log \frac{1}{q}.$$

For a random variable X with probability law μ , the positive-sign Fourier–Stieltjes transform of μ is its characteristic function, $\varphi_X(t) = \varphi_\mu(t) = \mathbb{E}e^{itX}$.

Definition 2.1 (Jacobi digit and Jacobi-digit cascade). Let U_α have density

$$\rho_\alpha(u) = c_\alpha(1 - u^2)^{\alpha-1} \mathbf{1}_{|u|<1}, \quad c_\alpha := \frac{\Gamma(\alpha + \frac{1}{2})}{\sqrt{\pi} \Gamma(\alpha)}. \quad (1)$$

For independent copies $U_{\alpha,0}, U_{\alpha,1}, \dots$, define

$$X_{q,\alpha} := a \sum_{n=0}^{\infty} q^n U_{\alpha,n}. \quad (2)$$

Its law and density, when convenient, are denoted by $\mu_{q,\alpha}$ and $f_{q,\alpha}$.

The series converges absolutely almost surely and $|X_{q,\alpha}| \leq 1$.

Lemma 2.2 (Beta representation and digit moments). *The affine variable $V_\alpha = (1 - U_\alpha)/2$ has the $\text{Beta}(\alpha, \alpha)$ distribution. Moreover,*

$$\mathbb{E}U_\alpha^{2m} = \frac{(1/2)^{\overline{m}}}{(\alpha + 1/2)^{\overline{m}}}, \quad \mathbb{E}U_\alpha^{2m+1} = 0, \quad (3)$$

and, for $p > -1$,

$$\mathbb{E}|U_\alpha|^p = \frac{\Gamma((p+1)/2)\Gamma(\alpha+1/2)}{\sqrt{\pi}\Gamma(\alpha+(p+1)/2)}. \quad (4)$$

In particular, $\text{Var}(U_\alpha) = 1/(2\alpha + 1)$.

Proof. The normalization integral is

$$\int_{-1}^1 (1 - u^2)^{\alpha-1} du = B\left(\frac{1}{2}, \alpha\right).$$

The substitution $w = (1 - u)/2$ gives the symmetric beta density. For even moments, symmetry and $s = u^2$ yield

$$2c_\alpha \int_0^1 u^{2m} (1 - u^2)^{\alpha-1} du = c_\alpha B\left(m + \frac{1}{2}, \alpha\right),$$

which simplifies to (3); the same calculation with $|u|^p$ proves (4). $\square$

Proposition 2.3 (Self-similarity). *With $X'_{q,\alpha}$ an independent copy of $X_{q,\alpha}$,*

$$X_{q,\alpha} \stackrel{d}{=} aU_\alpha + qX'_{q,\alpha}. \quad (5)$$

This fixed point is unique among probability laws on $[-1, 1]$.

Proof. Split the first term from (2). Conversely, repeated substitution of a solution of (5) gives the first N digits plus $q^{N+1}X^{(N+1)}$; the last term converges uniformly to zero. Hence all finite-dimensional characteristic functions, and thus the law, are forced. $\square$

2.1 Sphere dimension and compact random flights

The exponent in (1) has a direct geometric meaning.

Theorem 2.4 (Spherical random-flight realization). *Let $d \geq 2$ be an integer and put $\alpha = (d - 1)/2$. If $\Omega_0, \Omega_1, \dots$ are independent and uniform on the unit sphere $S^{d-1} \subset \mathbb{R}^d$, define*

$$\mathbf{X}_{q,d} := a \sum_{n \geq 0} q^n \Omega_n. \quad (6)$$

Then $\mathbf{X}_{q,d}$ is rotationally invariant, and every coordinate has the law of $X_{q,(d-1)/2}$. Its support is the closed annulus

$$\text{supp}(\mathbf{X}_{q,d}) = \left\{ x \in \mathbb{R}^d : \max(1 - 2q, 0) \leq \|x\| \leq 1 \right\}. \quad (7)$$

Consequently, the support changes from an annulus to the full unit ball at the critical scale $q = 1/2$.

Proof. The first coordinate of a uniform point on S^{d-1} has density proportional to $(1 - u^2)^{(d-3)/2}$, which is (1) with $\alpha = (d - 1)/2$. Rotational invariance is preserved by independent sums and weak limits.

For the support, write the step lengths as $b_n = aq^n$, so $\sum b_n = 1$ and the largest length is $b_0 = a$. In dimension at least two, the set of possible norms of a polygonal sum with prescribed lengths is the interval between the generalized polygon lower bound $\max(b_0 - \sum_{n \geq 1} b_n, 0)$ and the aligned upper bound $\sum b_n$. Finite truncations give every intermediate norm by continuously changing one angle, and passage to the limit preserves the interval. Since every open set of directions has positive product probability, the deterministic attainable set equals the probabilistic support. The lower endpoint is $\max(a - q, 0) = \max(1 - 2q, 0)$. $\square$

The limit $d \downarrow 1$, or $\alpha \downarrow 0$, is the sphere $S^0 = \{-1, +1\}$ and hence the Bernoulli-convolution boundary developed in Section 8. Thus $d = 1, 2, 3, \dots$ gives the sequence Rademacher, arcsine, uniform, semicircle-type, and higher spherical-coordinate digits; $d = 3$ is precisely the Rvachev digit.

Proposition 2.5 (First radial moments). *For $d \geq 2$, let*

$$S_2 = \sum_{n \geq 0} a^2 q^{2n} = \frac{a}{1+q}, \quad S_4 = \sum_{n \geq 0} a^4 q^{4n} = \frac{a^4}{1-q^4}.$$

Then

$$\mathbb{E} \|\mathbf{X}_{q,d}\|^2 = S_2, \quad (8)$$

$$\mathbb{E} \|\mathbf{X}_{q,d}\|^4 = S_2^2 + \frac{2}{d}(S_2^2 - S_4). \quad (9)$$

In particular,

$$\text{Var}(X_{q,\alpha}) = \frac{1-q}{(1+q)(2\alpha+1)}. \quad (10)$$

Proof. The mixed scalar products have zero expectation. On squaring $\|\sum b_n \Omega_n\|^2$, only identical unordered pairs survive in the fourth moment, and $\mathbb{E}(\Omega_i \cdot \Omega_j)^2 = 1/d$. This gives

$$S_2^2 + 4d^{-1} \sum_{i < j} b_i^2 b_j^2 = S_2^2 + 2d^{-1}(S_2^2 - S_4).$$

The coordinate variance is d^{-1} of the radial second moment. $\square$

3 Geometric Bessel products and smooth compact support

Define the normalized Bessel function

$$\mathcal{J}_\nu(z) := 2^\nu \Gamma(\nu+1) z^{-\nu} J_\nu(z) = {}_0F_1\left(\nu+1; -\frac{z^2}{4}\right), \quad (11)$$

where the removable value at $z = 0$ is one. It is even and entire. For $\nu = 1/2$, $\mathcal{J}_{1/2}(z) = \sin z/z$; for $\nu = -1/2$, $\mathcal{J}_{-1/2}(z) = \cos z$.

Theorem 3.1 (Geometric Bessel product). *The characteristic functions of the digit and cascade are*

$$\varphi_{U_\alpha}(t) = \mathcal{J}_{\alpha-1/2}(t), \quad (12)$$

$$\varphi_{X_{q,\alpha}}(t) = \prod_{n=0}^{\infty} \mathcal{J}_{\alpha-1/2}(aq^n t). \quad (13)$$

The product converges normally on compact subsets of $\mathbb{C}$ and is the unique entire solution of the Mahler-type equation

$$\Phi(t) = \mathcal{J}_\nu(at)\Phi(qt), \quad \Phi(0) = 1. \quad (14)$$

At $\alpha = 1$, this is the geometric sinc product already present in the repository; at $(q, \alpha) = (1/2, 1)$, it is the Fourier transform of the normalized Rvachev up density.

Proof. The classical beta–Bessel integral [4, 10] gives (12); it can also be verified by comparing the moment expansion in Theorem 2.2 with ${}_0F_1$. Independence gives finite products, and the tail converges normally because $\mathcal{J}_\nu(z) = 1 - z^2/[4(\nu + 1)] + O(z^4)$. Equation (14) follows by separating the first factor. Iterating it and using $\Phi(q^N t) \rightarrow \Phi(0) = 1$ proves uniqueness. $\square$

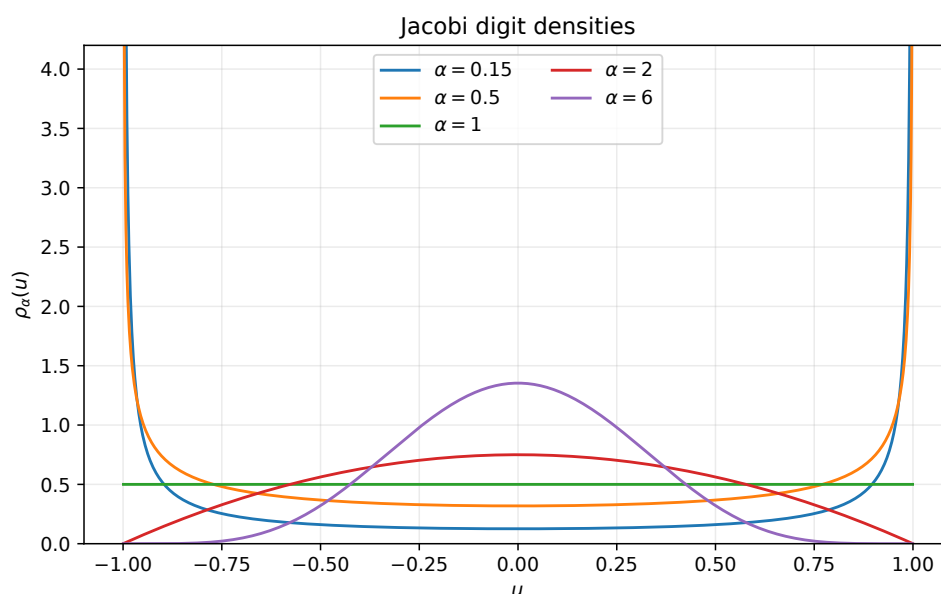


Figure 1: The one-step Jacobi digit. The limits $\alpha \downarrow 0$ and $\alpha \rightarrow \infty$ concentrate at the endpoints and at the origin, respectively; standardization of the latter gives a Gaussian.

Theorem 3.2 (Fourier decay, smoothness, and positivity). *For each fixed $q \in (0, 1)$ and $\alpha > 0$,*

$$\log |\varphi_{X_{q,\alpha}}(t)| \leq -\frac{\alpha}{2L} (\log |t|)^2 + O_{q,\alpha}(\log |t|) \quad (|t| \rightarrow \infty) \quad (15)$$

away from its zeros, with the left side interpreted as $-\infty$ at a zero. Consequently,

$$f_{q,\alpha} \in C_c^\infty(\mathbb{R}), \quad \text{supp}(f_{q,\alpha}) = [-1, 1],$$

and $f_{q,\alpha}(x) > 0$ for $-1 < x < 1$. Every derivative is flat at both endpoints.

Proof. The large-argument Bessel estimate gives $|\mathcal{J}_\nu(s)| \leq C_\alpha |s|^{-\alpha}$ for $|s| \geq 1$, while a characteristic function always has modulus at most one. Put $T = \log(a|t|)$ and sum this estimate over the indices $0 \leq n \leq T/L + O(1)$. The logarithm of the product is at most

$$O(T) - \alpha \sum_{n \leq T/L} (T - nL) = -\frac{\alpha}{2L} T^2 + O(T),$$

which is (15). The transform therefore decays faster than every power, so Fourier inversion gives a C^∞ density. The random series gives support contained in $[-1, 1]$, while the support of an infinite convolution is the closed Minkowski sum of the digit supports, which is the whole interval.

For positivity, write $X = aU + qX'$. At an interior point x , the intervals $(-q, q)$ and $(x - a, x + a)$ have nonempty open intersection. The law of qX' assigns positive mass to every open subinterval of its support, and the scaled digit kernel is positive on $(-a, a)$; their convolution is thus positive at x . Compact support plus smoothness implies endpoint flatness. $\square$

The same argument in $\mathbb{R}^d$ proves that the spherical random-flight law in Theorem 2.4 has a C^∞ density on its annular or ball support for every $d \geq 2$.

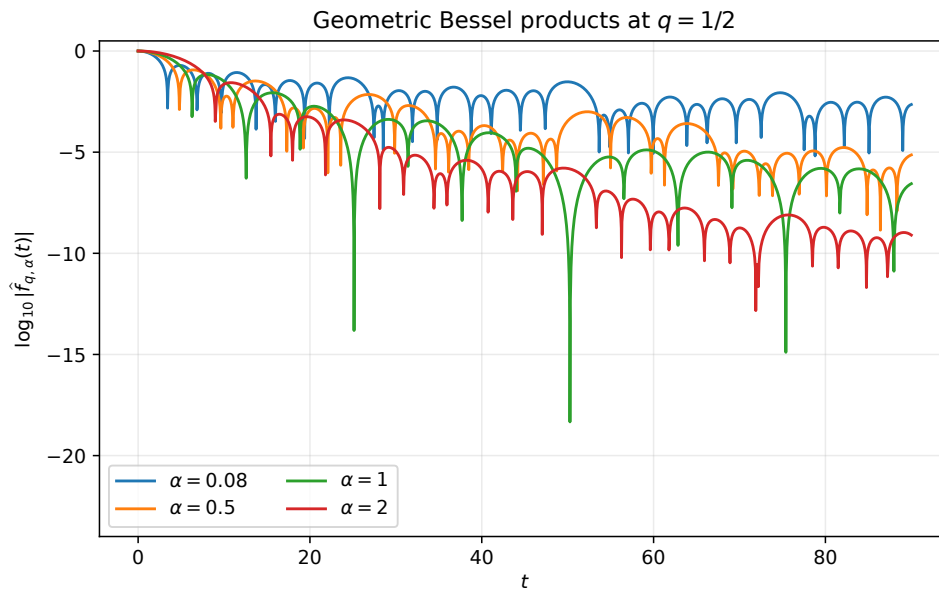


Figure 2: Logarithmic modulus of the geometric Bessel product at $q = 1/2$. The zeros are inherited from geometrically rescaled Bessel zeros, while the envelope obeys the quadratic logarithmic decay in (15).

Theorem 3.3 (Log-concavity above the Rvachev parameter). *For every $q \in (0, 1)$ and $\alpha \geq 1$, the density $f_{q, \alpha}$ is log-concave on $\mathbb{R}$. Hence it is unimodal and its superlevel sets are intervals.*

Proof. On $(-1, 1)$,

$$\frac{d^2}{du^2} \log \rho_\alpha(u) = -2(\alpha - 1) \frac{1 + u^2}{(1 - u^2)^2} \leq 0.$$

Together with the convex support interval, this makes the extended digit density log-concave. Scaling and convolution preserve log-concavity by the Prékopa–Leindler theorem [15]. Finite convolutions converge weakly to the cascade, and the limiting smooth density inherits log-concavity. $\square$

For $0 < \alpha < 1$, the digit is U-shaped and log-concavity is no longer automatic; numerical evidence in Section 11 suggests a nontrivial shape phase diagram.

3.1 A geometric q -Bessel coefficient array

Expanding every ${}_0F_1$ factor gives an exact positive formula for all even moments.

Theorem 3.4 (Occupation-sum moment formula). *For $N \geq 0$,*

$$\frac{\mathbb{E} X_{q, \alpha}^{2N}}{(2N)!} = \left(\frac{a^2}{4} \right)^N \sum_{\substack{k_0, k_1, \dots \geq 0 \\ k_0 + k_1 + \dots = N}} \prod_{n=0}^{\infty} \frac{q^{2nk_n}}{k_n! (\alpha + 1/2)^{k_n}}. \quad (16)$$

Only finitely many k_n are nonzero in each summand. Equivalently,

$$\sum_{N \geq 0} \frac{\mathbb{E} X_{q,\alpha}^{2N}}{(2N)!} t^{2N} = \prod_{n \geq 0} {}_0F_1 \left(; \alpha + \frac{1}{2}; \frac{a^2 q^{2n} t^2}{4} \right). \quad (17)$$

Proof. Use the defining series for ${}_0F_1$ in the moment-generating version of (13), multiply the absolutely convergent products, and collect the total degree $N = \sum k_n$. $\square$

Formula (16) is not one of the standard Jackson or Hahn–Exton q -Bessel functions; the q -dependence lies in a geometric cascade of classical Bessel atoms. It nevertheless supplies a natural q -Bessel coefficient array tailored to the Fabius–Rvachev refinement operator.

4 Rayleigh cumulants, Bell moments, and Bernoulli specializations

Let $j_{\nu,1} < j_{\nu,2} < \dots$ be the positive zeros of J_ν , and define the Rayleigh sums

$$R_m(\nu) := \sum_{k=1}^{\infty} j_{\nu,k}^{-2m}. \quad (18)$$

The classical Bessel product is

$$\mathcal{J}_\nu(z) = \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{j_{\nu,k}^2} \right). \quad (19)$$

Lemma 4.1 (Rayleigh recurrence). *For $\nu > -1$,*

$$R_1(\nu) = \frac{1}{4(\nu+1)}, \quad R_m(\nu) = \frac{1}{\nu+m} \sum_{r=1}^{m-1} R_r(\nu) R_{m-r}(\nu), \quad m \geq 2. \quad (20)$$

Proof. The normalized Bessel function $y = \mathcal{J}_\nu$ satisfies $y'' + (2\nu+1)y'/z + y = 0$. From (19),

$$w(z) := -\frac{y'(z)}{y(z)} = 2 \sum_{m \geq 1} R_m(\nu) z^{2m-1}.$$

The Bessel equation becomes the Riccati equation $w' = w^2 - (2\nu+1)w/z + 1$. Comparing coefficients gives (20); this is Kishore’s classical recurrence [6, 7]. $\square$

Theorem 4.2 (Rayleigh-factorized cumulants). *All odd cumulants of $X_{q,\alpha}$ vanish, and*

$$\kappa_{2m}(X_{q,\alpha}) = (-1)^{m+1} \frac{(2m)!}{m} R_m \left(\alpha - \frac{1}{2} \right) \frac{(1-q)^{2m}}{1-q^{2m}}. \quad (21)$$

If $[s]_q = (1-q^s)/(1-q)$, the scale factor is

$$\frac{(1-q)^{2m}}{1-q^{2m}} = \frac{(1-q)^{2m-1}}{[2m]_q}. \quad (22)$$

Thus the atom geometry and the geometric scale separate exactly into a Rayleigh factor and a q -integer factor.

Proof. Taking a logarithm in (19) gives, near zero,

$$\log \mathcal{J}_\nu(t) = - \sum_{m \geq 1} \frac{R_m(\nu)}{m} t^{2m}.$$

Comparison with $\log \mathbb{E} e^{itU} = \sum_r \kappa_r(U) (it)^r / r!$ yields $\kappa_{2m}(U) = (-1)^{m+1} (2m)! R_m / m$. Cumulants add over independent summands and scale by the corresponding power, so summing $a^{2m} q^{2mn}$ proves (21). $\square$

The first five nonzero digit cumulants, generated both from (20) and independently from the symbolic ${}_0F_1$ logarithm, are

$$\kappa_2(U_\alpha) = \frac{1}{2\alpha + 1}, \quad (23)$$

$$\kappa_4(U_\alpha) = -\frac{6}{(2\alpha + 1)^2(2\alpha + 3)}, \quad (24)$$

$$\kappa_6(U_\alpha) = \frac{240}{(2\alpha + 1)^3(2\alpha + 3)(2\alpha + 5)}, \quad (25)$$

$$\kappa_8(U_\alpha) = -\frac{5040(10\alpha + 17)}{(2\alpha + 1)^4(2\alpha + 3)^2(2\alpha + 5)(2\alpha + 7)}, \quad (26)$$

$$\kappa_{10}(U_\alpha) = \frac{725760(14\alpha + 31)}{(2\alpha + 1)^5(2\alpha + 3)^2(2\alpha + 5)(2\alpha + 7)(2\alpha + 9)}. \quad (27)$$

Corollary 4.3 (Bell-polynomial moments). *Let B_n denote the complete exponential Bell polynomial. Then*

$$\mathbb{E}X_{q,\alpha}^n = B_n(0, \kappa_2, 0, \kappa_4, \dots, \kappa_n), \quad (28)$$

with the cumulants from (21). Equivalently,

$$\mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}, \quad \mu_0 = 1. \quad (29)$$

The distributional fixed point also gives the positive recurrence

$$(1 - q^{2N})\mu_{2N} = \sum_{r=1}^N \binom{2N}{2r} a^{2r} q^{2N-2r} \frac{(1/2)^{\bar{r}}}{(\alpha + 1/2)^{\bar{r}}} \mu_{2N-2r}. \quad (30)$$

Proof. Equation (28) is the standard moment–cumulant relation and (29) is its differentiated generating-function form. Expanding $(aU + qX')^{2N}$, using symmetry and independence, and moving the term with $r = 0$ to the left proves (30). $\square$

4.1 Bernoulli numbers at two different boundaries

At $\alpha = 1$, $j_{1/2,k} = k\pi$, hence $R_m(1/2) = \zeta(2m)/\pi^{2m}$. Euler’s evaluation of $\zeta(2m)$ gives the familiar uniform-digit cumulants

$$\kappa_{2m}(U_1) = \frac{2^{2m-1}B_{2m}}{m}, \quad \kappa_{2m}(X_{q,1}) = \frac{2^{2m-1}B_{2m}}{m} \frac{a^{2m}}{1 - q^{2m}}. \quad (31)$$

This is an established repository specialization.

At the opposite limit $\alpha \downarrow 0$, the digit becomes a Rademacher variable, whose cumulants are

$$\kappa_{2m}(\varepsilon) = \frac{2^{2m}(2^{2m} - 1)B_{2m}}{2m}. \quad (32)$$

Thus Bernoulli numbers occur at both the uniform/Rvachev slice and the Bernoulli-convolution boundary, but with different multipliers. The Rayleigh recurrence analytically interpolates between them.

5 Fourier zeros, spectral zeta, and scale resonance

5.1 Exact zero multiset and zeta factorization

By (13), the positive Fourier-zero multiset is

$$\Lambda_{q,\alpha} = \left\{ \lambda_{n,k} := \frac{j_{\nu,k}}{aq^n} : n \geq 0, k \geq 1 \right\}, \quad (33)$$

where coincidences are counted with multiplicity.

Theorem 5.1 (Geometric Bessel spectral zeta). *For $\Re s > 1$, let*

$$\zeta_\nu^B(s) = \sum_{k \geq 1} j_{\nu,k}^{-s}, \quad \mathcal{Z}_{q,\alpha}(s) = \sum_{n,k} \lambda_{n,k}^{-s}. \quad (34)$$

Then

$$\mathcal{Z}_{q,\alpha}(s) = \frac{(1-q)^s}{1-q^s} \zeta_{\alpha-1/2}^B(s) = \frac{(1-q)^{s-1}}{[s]_q} \zeta_{\alpha-1/2}^B(s). \quad (35)$$

At the even positive integers,

$$\kappa_{2m}(X_{q,\alpha}) = (-1)^{m+1} \frac{(2m)!}{m} \mathcal{Z}_{q,\alpha}(2m). \quad (36)$$

Proof. Insert (33), separate the two absolutely convergent sums, and evaluate $\sum_{n \geq 0} q^{ns}$. Equation (36) is (21). $\square$

Under the standard meromorphic continuation of the Bessel zeta, the geometric factor adds the lattice of possible poles

$$\chi_\ell = \frac{2\pi i \ell}{L}, \quad \ell \in \mathbb{Z}, \quad (37)$$

subject to cancellations by ζ_ν^B . At a noncancelled pole,

$$\operatorname{Res}_{s=\chi_\ell} \mathcal{Z}_{q,\alpha}(s) = \frac{a^{\chi_\ell} \zeta_\nu^B(\chi_\ell)}{L}. \quad (38)$$

These are the spectral counterpart of the log-periodic endpoint term proved in Theorem 7.3.

5.2 Resonance is exceptional but dense

Definition 5.2 (Resonant scales). For fixed $\nu > -1/2$, define

$$\mathcal{E}_\nu := \left\{ \left(\frac{j_{\nu,k}}{j_{\nu,\ell}} \right)^{1/d} : 1 \leq k < \ell, d \in \mathbb{N}_0 \right\} \subset (0, 1). \quad (39)$$

Theorem 5.3 (Countable-dense resonance dichotomy). *The set $\mathcal{E}_\nu$ is countable and dense in $(0, 1)$. If $q \notin \mathcal{E}_\nu$, every nonzero zero of $\varphi_{X_{q,\alpha}}$ is simple. If $q \in \mathcal{E}_\nu$, at least one zero has multiplicity at least two. Hence simple zeros occur for Lebesgue almost every q , even though resonant q 's are dense.*

Proof. Two factors share a positive zero precisely when

$$\frac{j_{\nu,\ell}}{aq^n} = \frac{j_{\nu,k}}{aq^{n+d}}, \quad d \geq 1,$$

which is equivalent to $q^d = j_{\nu,k}/j_{\nu,\ell}$. Bessel zeros are simple, so absence of such an equality is equivalent to simplicity of the full product zeros. Countability is immediate.

For density it suffices to use $d = 1$. McMahon's asymptotic formula [5, 4] $j_{\nu,k} = \pi(k + \nu/2 - 1/4) + O(k^{-1})$ implies that, whenever $k_\ell/\ell \rightarrow r \in (0, 1)$, $j_{\nu,k_\ell}/j_{\nu,\ell} \rightarrow r$. Choosing $k_\ell = \lfloor r\ell \rfloor$ proves density. $\square$

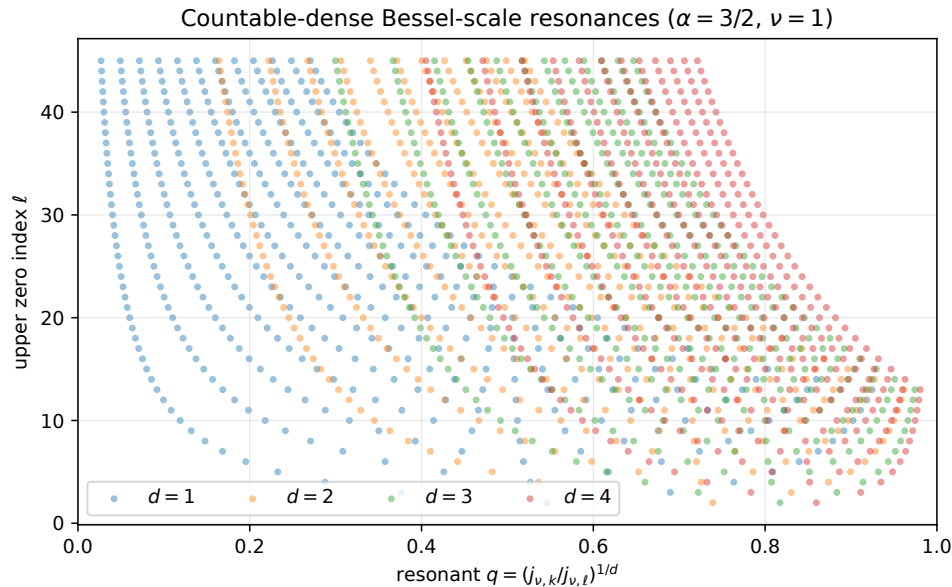


Figure 3: A finite sample of the countable-dense resonance set for $\alpha = 3/2$, so $\nu = 1$. The four point families correspond to scale separations $d = 1, 2, 3, 4$.

The maximum gap among the ratios $j_{1,k}/j_{1,\ell}$, augmented by zero and one, decreases as follows:

number of zeros	number of ratios	maximum gap	median gap
10	45	0.11903523	0.01781670
20	190	0.06023619	0.00429132
40	780	0.03030307	0.00104959
80	3160	0.01519847	0.00026183

This table is illustrative, not needed for the density proof.

Corollary 5.4 (Sign from zero counting). *Away from its zeros,*

$$\operatorname{sgn} \varphi_{X_{q,\alpha}}(t) = (-1)^{N_{q,\alpha}(|t|)}, \quad (40)$$

where $N_{q,\alpha}(R)$ counts positive zeros at most R , with multiplicity. At $\alpha = 1, q = 1/2$, arithmetic coincidences among $j_{1/2,k} = k\pi$ reduce this parity to the dyadic Thue–Morse law already proved in the repository. For generic α , the ordering is governed instead by Bessel-zero ratios.

Theorem 5.5 (Universal zero-density law). *Counting multiplicity,*

$$N_{q,\alpha}(R) = \frac{R}{\pi} + O_{q,\alpha}(\log R) \quad (R \rightarrow \infty). \quad (41)$$

The leading density $1/\pi$ is independent of both q and α .

Proof. The Bessel zero count satisfies $N_\nu(x) = x/\pi + O_\nu(1)$. Therefore

$$N_{q,\alpha}(R) = \sum_{n \geq 0} N_\nu(aq^n R).$$

Only $O(\log R)$ summands are nonzero. The main terms sum to

$$\frac{aR}{\pi} \sum_{n \geq 0} q^n = \frac{R}{\pi},$$

and truncating this geometric sum where $aq^n R \asymp 1$ changes it by $O(1)$. The bounded error in each active level contributes $O(\log R)$. $\square$

For the sinc slice, the $O(\log R)$ term contains digital sums. For generic Bessel zeros it becomes a quasidigital fluctuation generated by the orbit of $\log R$ modulo L . A smoothed counting function should admit a full Fourier expansion from the residues (38); see Theorem 12.2.

6 Higher-order Rvachev refinement equations

The ordinary Rvachev law is distinguished by a uniform one-step kernel, whose first distributional derivative consists of two endpoint delta masses. For integer $\alpha = m$, the Jacobi kernel is a compact polynomial and the same mechanism survives at order $2m - 1$.

Let

$$k_{a,m}(x) = c_m a^{-(2m-1)} (a^2 - x^2)_+^{m-1}, \quad c_m = \frac{\Gamma(m+1/2)}{\sqrt{\pi} \Gamma(m)}, \quad (42)$$

and $g(x) = q^{-1} f_{q,m}(x/q)$. The fixed point is $f_{q,m} = k_{a,m} * g$.

Theorem 6.1 (Finite higher-order refinement). *For every integer $m \geq 1$, in the classical sense on $\mathbb{R}$,*

$$\begin{aligned} f_{q,m}^{(2m-1)}(x) = & (-1)^{m-1} c_m a^{-(2m-1)} \sum_{j=0}^{m-1} \frac{(2m-2-j)!(m-1)!}{(m-1-j)!j!} (2a)^j q^{-j-1} \\ & \times \left[(-1)^j f_{q,m}^{(j)}\left(\frac{x+a}{q}\right) - f_{q,m}^{(j)}\left(\frac{x-a}{q}\right) \right]. \end{aligned} \quad (43)$$

Thus a compact polynomial digit of degree $2m - 2$ produces a finite functional-differential equation of order $2m - 1$.

Proof. Put $P(x) = (a^2 - x^2)^{m-1}$ on $[-a, a]$ and zero outside. Its $(2m - 1)$ -st distributional derivative is supported at the endpoints. The jump formula gives

$$D^{2m-1} P_+ = \sum_{j=0}^{m-1} P^{(2m-2-j)}(a) ((-1)^j \delta_{-a}^{(j)} - \delta_a^{(j)}).$$

Only $j \leq m - 1$ survives because P has a zero of order $m - 1$ at each endpoint. Leibniz' rule gives

$$P^{(2m-2-j)}(a) = (-1)^{m-1} \frac{(2m-2-j)!(m-1)!}{(m-1-j)!j!} (2a)^j.$$

Convolution with g , followed by $g^{(j)}(x) = q^{-j-1} f^{(j)}(x/q)$, proves (43). The distributional identity is classical because $f \in C_c^\infty$. $\square$

The first three equations are instructive. For $m = 1$,

$$f'_{q,1}(x) = \frac{1}{2aq} \left[f_{q,1}\left(\frac{x+a}{q}\right) - f_{q,1}\left(\frac{x-a}{q}\right) \right]. \quad (44)$$

At $q = a = 1/2$, this is $\text{up}'(x) = 2[\text{up}(2x+1) - \text{up}(2x-1)]$. For $m = 2$,

$$\begin{aligned} f'''_{q,2}(x) = & -\frac{3}{2a^3q} \left[f_{q,2}\left(\frac{x+a}{q}\right) - f_{q,2}\left(\frac{x-a}{q}\right) \right] \\ & + \frac{3}{2a^2q^2} \left[f'_{q,2}\left(\frac{x+a}{q}\right) + f'_{q,2}\left(\frac{x-a}{q}\right) \right]. \end{aligned} \quad (45)$$

For $m = 3$,

$$f^{(5)}_{q,3}(x) = \frac{45}{2a^5q} [f_+ - f_-] - \frac{45}{2a^4q^2} [f'_+ + f'_-] + \frac{15}{2a^3q^3} [f''_+ - f''_-], \quad (46)$$

where $f_{\pm}^{(j)} = f_{q,3}^{(j)}((x \pm a)/q)$.

Equation (43) suggests replacing the scalar Thue–Morse sign cocycle of $m = 1$ by a finite-dimensional jet cocycle for $m > 1$. This is made precise as a research problem in Theorem 12.3.

7 Endpoint fractional pantographs and inverse asymptotics

Let

$$D_{q,\alpha} := 1 - X_{q,\alpha} \in [0, 2], \quad G_{q,\alpha}(x) := \mathbb{P}(D_{q,\alpha} \leq x). \quad (47)$$

Since $V_\alpha = (1 - U_\alpha)/2 \sim \text{Beta}(\alpha, \alpha)$,

$$D_{q,\alpha} \stackrel{d}{=} 2aV_\alpha + qD'_{q,\alpha}. \quad (48)$$

7.1 Exact Volterra equation

Proposition 7.1 (Endpoint Volterra–pantograph equation). *For $0 < x < 2a$,*

$$G(x) = \frac{1}{(2a)^\alpha B(\alpha, \alpha)} \int_0^x (x-y)^{\alpha-1} \left(1 - \frac{x-y}{2a}\right)^{\alpha-1} G\left(\frac{y}{q}\right) dy. \quad (49)$$

Proof. Condition on $2aV_\alpha = t$ in (48) and then change variables from t to $y = x - t$. □

For $\beta > 0$, write the left Riemann–Liouville integral as

$$(I^\beta h)(x) = \frac{1}{\Gamma(\beta)} \int_0^x (x-y)^{\beta-1} h(y) dy. \quad (50)$$

Because G is flat at zero, the corresponding Riemann–Liouville derivative D^α is an unambiguous left inverse of I^α here.

Theorem 7.2 (Exact fractional endpoint equation). *For $0 < x < 2a$,*

$$D^\alpha G(x) = \frac{\Gamma(2\alpha)}{\Gamma(\alpha)(2a)^\alpha} \sum_{r=0}^{\infty} \frac{(\alpha)^{\bar{r}} (1-\alpha)^{\bar{r}}}{r!(2a)^r} I^r \left[G\left(\frac{\cdot}{q}\right) \right] (x). \quad (51)$$

The series converges locally absolutely for $x < 2a$. If $\alpha = m \in \mathbb{N}_0$, it terminates after $r = m - 1$. For $\alpha = 1$,

$$G'(x) = \frac{1}{2a} G(x/q), \quad 0 < x < 2a. \quad (52)$$

Proof. Expand the second factor in (49):

$$\left(1 - \frac{x-y}{2a}\right)^{\alpha-1} = \sum_{r \geq 0} \frac{(1-\alpha)^{\bar{r}}}{r!(2a)^r} (x-y)^r.$$

On compact subintervals of $(0, 2a)$, termwise integration is legitimate. The integral in the r -th term is $\Gamma(\alpha + r) I^{\alpha+r} [G(\cdot/q)]$. Apply D^α , use $\Gamma(\alpha + r) = \Gamma(\alpha) (\alpha)^{\bar{r}}$, and simplify the beta constant. For convergence, $I^r h = O(x^r/r!)$, while the Pochhammer coefficient grows only like $r!/r$ up to a power; the resulting radius is $2a$. If $\alpha = m$, $(1-m)^{\bar{r}} = 0$ for $r \geq m$. □

Under the common normalization $\text{Fab}(x) = \text{up}(x-1)$ on $[0, 1]$, symmetry gives $G'_{1/2,1}(x) = \text{Fab}(x)$, while (52) gives $G'_{1/2,1}(x) = G_{1/2,1}(2x)$. Hence

$$\text{Fab}(x) = G_{1/2,1}(2x), \quad 0 \leq x \leq \frac{1}{2}. \quad (53)$$

Thus (51) is a genuine fractional-order extension of the local Fabius equation, not merely an analogy.

7.2 Endpoint Laplace product and periodic renormalization

The beta Laplace transform is

$$M_\alpha(s) := \mathbb{E}e^{-sV_\alpha} = {}_1F_1(\alpha; 2\alpha; -s). \quad (54)$$

Therefore

$$\mathbb{E}e^{-sD_{q,\alpha}} = \prod_{n \geq 0} M_\alpha(2aq^n s). \quad (55)$$

Set $C_\alpha = \Gamma(2\alpha)/\Gamma(\alpha)$ and $h_\alpha(y) = \log M_\alpha(y)$. Standard Kummer asymptotics [4] give

$$h_\alpha(y) = \log C_\alpha - \alpha \log y + O(y^{-1}) \quad (y \rightarrow \infty), \quad h_\alpha(y) = O(y) \quad (y \downarrow 0). \quad (56)$$

Theorem 7.3 (Exact source of the log-periodic endpoint term). *Let*

$$\mathcal{K}_{\alpha,q}(z) = \sum_{n \geq 0} h_\alpha(zq^n).$$

There is a real-analytic, one-periodic function $P_{\alpha,q}$ such that

$$\mathcal{K}_{\alpha,q}(z) = -\frac{\alpha}{2L}(\log z)^2 + \left(\frac{\log C_\alpha}{L} - \frac{\alpha}{2}\right) \log z + P_{\alpha,q}\left(\frac{\log z}{L}\right) + O(z^{-1}) \quad (57)$$

as $z \rightarrow \infty$, where the argument of P is taken modulo one. An explicit convergent representation is obtained as follows. For $0 \leq \theta < 1$, put

$$\begin{aligned} P_{\alpha,q}(\theta) &= \sum_{m=-\infty}^{-1} h_\alpha(e^{L(m+\theta)}) + \sum_{m=0}^{\infty} [h_\alpha(e^{L(m+\theta)}) - \log C_\alpha + \alpha L(m+\theta)] \\ &\quad + \frac{\alpha L}{2}(\theta^2 - \theta) + (\log C_\alpha)(1 - \theta). \end{aligned} \quad (58)$$

Proof. Write $\log z = L(N + \theta)$, where $N = \lfloor \log z/L \rfloor$, and reindex $m = N - n$:

$$\mathcal{K}(z) = \sum_{m=-\infty}^N h_\alpha(e^{L(m+\theta)}).$$

For $m \geq 0$, add and subtract $\log C_\alpha - \alpha L(m + \theta)$. The negative tail converges because $h_\alpha(e^{Lm}) = O(e^{Lm})$; the corrected positive tail converges because of the $O(e^{-Lm})$ remainder in (56). Summing the subtracted arithmetic progression and replacing N by $\log z/L - \theta$ gives the quadratic and linear terms in (57), plus precisely the last line of (58). Reindexing proves one-periodicity. For real analyticity, complexify θ . The beta Laplace transform is zero-free in a fixed relative neighborhood of the positive axis for sufficiently small and sufficiently large arguments (by its Taylor and Kummer expansions), and compactness handles the intervening annulus. Hence the logarithms can be chosen in a common strip $|\Im \theta| < \eta$, where both tails converge normally. Their sum is holomorphic in that strip and therefore real analytic on the real axis. The omitted positive tail is $O(e^{-LN}) = O(z^{-1})$. $\square$

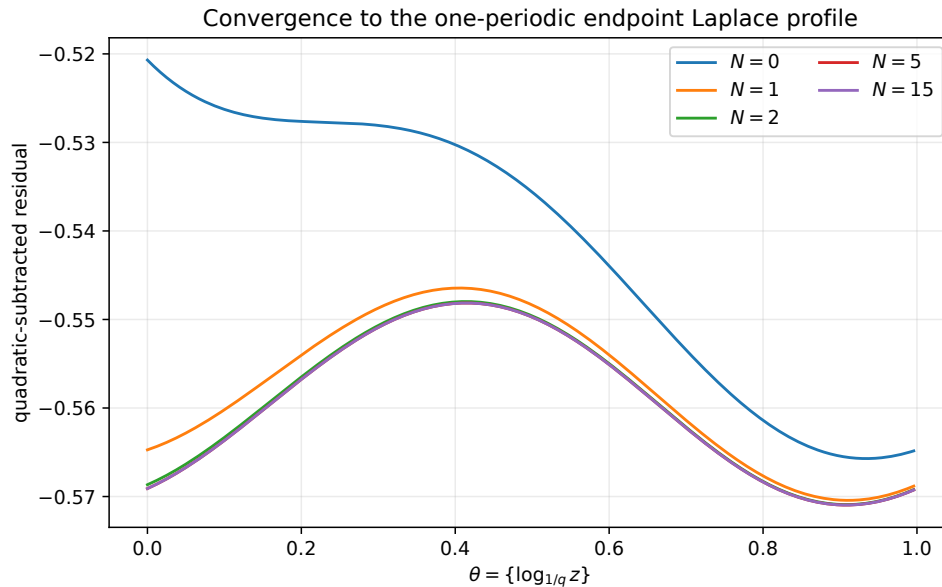


Figure 4: After subtraction of the quadratic and linear terms in (57), successive logarithmic decays converge rapidly to the same one-periodic profile. The plotted parameters are $\alpha = 1.3$, $q = 0.1$.

7.3 Small balls and inverse quantiles

Theorem 7.4 (Endpoint log-square law). As $x \downarrow 0$,

$$\lim_{x \downarrow 0} \frac{\log G_{q,\alpha}(x)}{(\log(1/x))^2} = -\frac{\alpha}{2 \log(1/q)}. \quad (59)$$

More quantitatively,

$$\log G_{q,\alpha}(x) = -\frac{\alpha}{2L} (\log(1/x))^2 + O_{q,\alpha}(\log(1/x) \log \log(1/x)). \quad (60)$$

Proof. Write $D = \sum b_n V_n$, where $b_n = 2aq^n$. Near zero, the beta CDF satisfies $c_1 y^\alpha \leq \mathbb{P}(V \leq y) \leq c_2 y^\alpha$.

For an upper bound, choose N so that $b_N \gtrsim x$. The event $D \leq x$ implies $V_n \leq x/b_n$ for $0 \leq n \leq N$. Independence gives

$$\log G(x) \leq \alpha \sum_{n=0}^N (\log x - \log b_n) + O(N).$$

Since $N = \log(1/x)/L + O(1)$, the arithmetic sum has leading term $-\alpha(\log(1/x))^2/(2L)$.

For a lower bound, choose N so the deterministic tail $\sum_{n>N} b_n = 2q^{N+1} \leq x/2$, and impose $b_n V_n \leq x/[2(N+1)]$ for every $n \leq N$. This event implies $D \leq x$, and its logarithmic probability is the same arithmetic sum with an additional $-\alpha(N+1) \log(2N+2) + O(N)$. The latter is $O(\log(1/x) \log \log(1/x))$, proving both claims. $\square$

Let $Q_{q,\alpha}(y) = G_{q,\alpha}^{-1}(y)$ be the lower endpoint quantile.

Corollary 7.5 (Inverse endpoint scale). As $y \downarrow 0$,

$$\log \frac{1}{Q_{q,\alpha}(y)} \sim \sqrt{\frac{2 \log(1/q)}{\alpha}} \log \frac{1}{y}. \quad (61)$$

For the usual Fabius normalization,

$$\text{Fab}^{-1}(y) = \frac{1}{2} \exp \left[-\sqrt{2 \log 2 \log(1/y)} (1 + o(1)) \right]. \quad (62)$$

Proof. Monotonic inversion of (59) gives (61). Equation (53) implies $\text{Fab}^{-1}(y) = Q_{1/2,1}(y)/2$ for small y . $\square$

Conjecture 7.6 (Periodic Lambert- W_{-1} transseries). *Let s_x be the positive saddle satisfying*

$$\frac{d}{ds} \log \mathbb{E} e^{-sD_{q,\alpha}} + x = 0, \quad T_x = \log(2as_x).$$

Then

$$T_x = -W_{-1} \left(-\frac{Lx}{2a\alpha} \right) + O(1), \quad (63)$$

and the complete endpoint expansion is obtained by steepest descent from

$$G(x) \sim \frac{\exp\{\mathcal{K}_{\alpha,q}(2as_x) + s_x x\}}{s_x \sqrt{2\pi \mathcal{K}_{\alpha,q}''(2as_x)(2a)^2}} \left(1 + \sum_{r \geq 1} \frac{c_r(\{T_x/L\})}{T_x^r} \right), \quad (64)$$

where every c_r is one-periodic and real analytic. Reversion of this series gives a complete inverse-quantile expansion. At $\alpha = 1, q = 1/2$, it should coincide term by term with the repository's all-orders inverse Fabius expansion after normalization changes.

The leading Lambert relation (with the branch conventions of [12]) follows formally by differentiating (57): $xs \sim \alpha T/L$, and hence $Te^{-T} \sim Lx/(2a\alpha)$. The unresolved part is a uniform saddle-point transfer retaining the periodic derivatives and controlling all remainder terms.

8 Boundary regimes: Bernoulli, Rvachev, and Gaussian

8.1 The Bernoulli-convolution boundary $\alpha \downarrow 0$

Let ε_n be independent Rademacher variables and

$$B_q = a \sum_{n \geq 0} q^n \varepsilon_n. \quad (65)$$

This is the normalized Bernoulli convolution with parameter q ; see, for context, [18, 19, 20].

Theorem 8.1 (Uniform Wasserstein desingularization rate). *For every $q \in (0, 1)$,*

$$W_1(\mu_{q,\alpha}, \text{Law}(B_q)) \leq 1 - \frac{\Gamma(\alpha + 1/2)}{\sqrt{\pi} \Gamma(\alpha + 1)} = 2\alpha \log 2 + O(\alpha^2) \quad (66)$$

as $\alpha \downarrow 0$. The bound is independent of q . Consequently, $X_{q,\alpha} \Rightarrow B_q$ uniformly in the scale parameter in the W_1 sense.

Proof. Couple $\varepsilon_n = \text{sgn}(U_{\alpha,n})$. By symmetry it is Rademacher, and

$$\mathbb{E}|X_{q,\alpha} - B_q| \leq a \sum_{n \geq 0} q^n \mathbb{E}(1 - |U_\alpha|) = 1 - \mathbb{E}|U_\alpha|.$$

Use (4) with $p = 1$. The expansion follows by differentiating the logarithm of the gamma ratio at zero and using $\psi(1/2) - \psi(1) = -2 \log 2$. $\square$

For every $\alpha > 0$, [Theorem 3.2](#) gives a positive C_c^∞ density. At $\alpha = 0$, however, the limit can be singular: for $q < 1/2$ it is the separated Cantor-type Bernoulli measure. Thus α is a genuine smooth desingularization parameter. At the critical value $q = 1/2$, the binary product identity

$$\prod_{n=0}^{\infty} \cos(2^{-n-1}t) = \frac{\sin t}{t} \quad (67)$$

shows that the $\alpha = 0$ limit is uniform on $[-1, 1]$, while $\alpha = 1$ is the Rvachev up law.

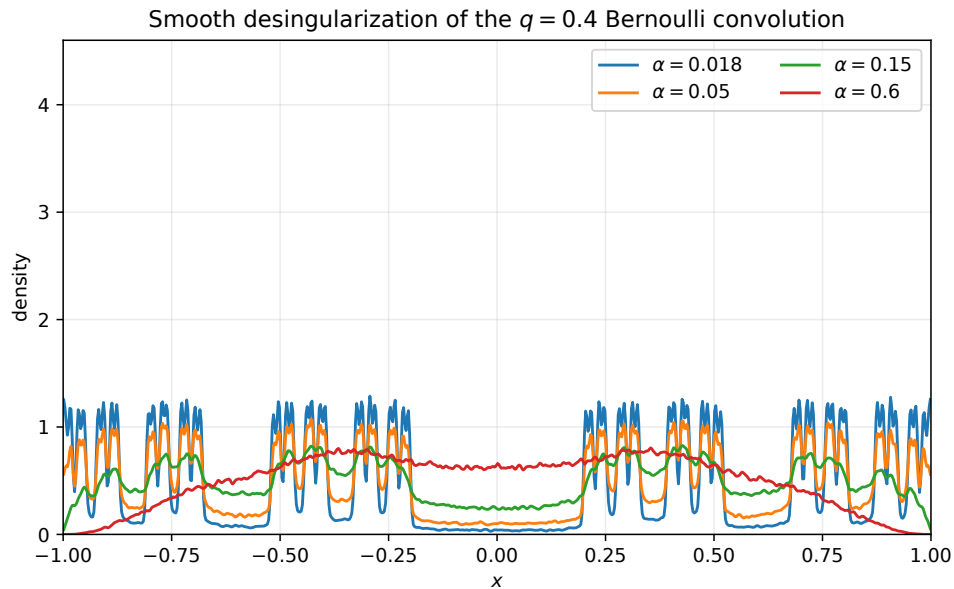


Figure 5: For $q = 0.4 < 1/2$, small positive α gives a smooth density that increasingly resolves the gaps and peaks of the limiting Cantor-type Bernoulli convolution. The figure is a fixed-seed histogram and is used only for qualitative shape evidence.

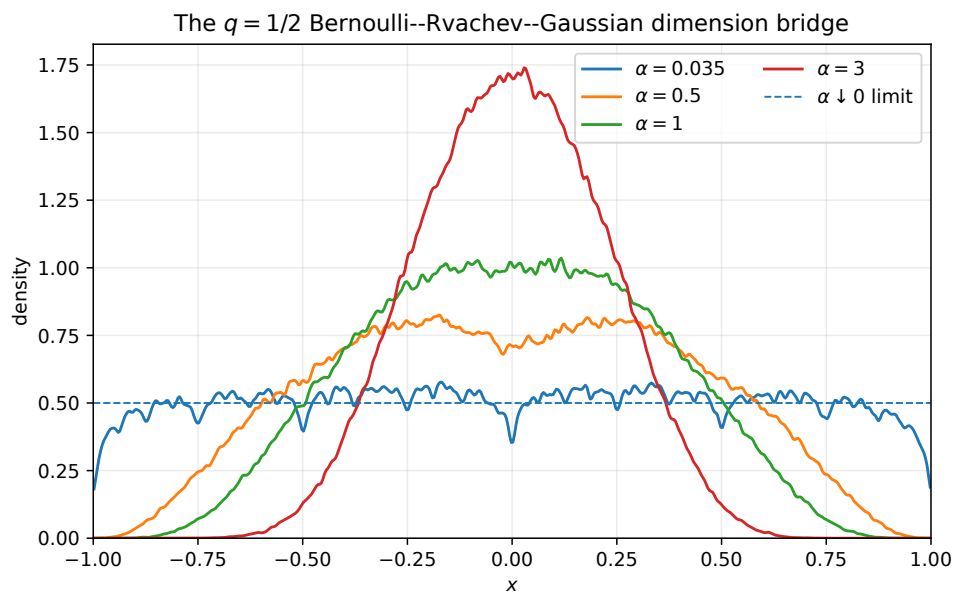


Figure 6: At $q = 1/2$, $\alpha \downarrow 0$ tends to the uniform density; $\alpha = 1$ is the Rvachev law; increasing α produces an increasingly Gaussian central profile after the natural variance change.

8.2 Gaussianization as $q \uparrow 1$

Let $Z_{q,\alpha} = X_{q,\alpha}/\sigma_{q,\alpha}$, with variance from (10). The standardized cumulants factor as

$$\lambda_{2m}(q, \alpha) := \frac{\kappa_{2m}(X_{q,\alpha})}{\sigma_{q,\alpha}^{2m}} = \lambda_{2m}(U_\alpha) \frac{(1 - q^2)^m}{1 - q^{2m}}. \quad (68)$$

In particular,

$$\lambda_4(q, \alpha) = -\frac{6}{2\alpha + 3} \frac{1 - q^2}{1 + q^2}, \quad (69)$$

$$\lambda_6(q, \alpha) = \frac{240}{(2\alpha + 3)(2\alpha + 5)} \frac{(1 - q^2)^2}{1 + q^2 + q^4}. \quad (70)$$

Theorem 8.2 (Central limit and Berry–Esseen scale). *For fixed $\alpha > 0$,*

$$Z_{q,\alpha} \Rightarrow N(0, 1) \quad (q \uparrow 1). \quad (71)$$

Moreover, with C_{BE} any valid universal Berry–Esseen constant,

$$d_K(Z_{q,\alpha}, N(0, 1)) \leq C_{\text{BE}} \rho_3(\alpha) \frac{(1 - q^2)^{3/2}}{1 - q^3}, \quad (72)$$

where

$$\rho_3(\alpha) = (2\alpha + 1)^{3/2} \frac{\Gamma(\alpha + 1/2)}{\sqrt{\pi} \Gamma(\alpha + 2)}. \quad (73)$$

The right side is $O_\alpha(\sqrt{1 - q})$.

Proof. Apply the Berry–Esseen theorem [16, 17] to finite truncations of the independent weighted sum and pass to the limit. The ratio of the weighted third absolute moment to the variance to the power $3/2$ is

$$\frac{\sum a^3 q^{3n} \mathbb{E}|U|^3}{(\sum a^2 q^{2n} \text{Var } U)^{3/2}} = \rho_3(\alpha) \frac{(1 - q^2)^{3/2}}{1 - q^3}.$$

The asymptotic order follows by expansion at $q = 1$. $\square$

8.3 Gaussianization as $\alpha \rightarrow \infty$

For fixed q , (68) and (24)–(27) show

$$\lambda_{2m}(q, \alpha) = O_{m,q}(\alpha^{1-m}), \quad m \geq 2. \quad (74)$$

Moment convergence therefore gives a second Gaussian route.

Theorem 8.3 (Two-parameter Gaussian corner). *If either $q \uparrow 1$ with α bounded away from zero, or $\alpha \rightarrow \infty$ with q in a compact subset of $(0, 1)$, then $Z_{q,\alpha} \Rightarrow N(0, 1)$. In particular, the conclusion holds along every path with $(1 - q) + \alpha^{-1} \rightarrow 0$. On bounded Fourier intervals,*

$$\log \mathbb{E} e^{itZ_{q,\alpha}} = -\frac{t^2}{2} + \frac{\lambda_4 t^4}{24} + O(t^6 \lambda_6 + t^8 \lambda_4^2). \quad (75)$$

At the level of the first formal Edgeworth approximation, the density correction is

$$\varphi(z) \left[1 + \frac{\lambda_4}{24} H_4(z) \right], \quad H_4(z) = z^4 - 6z^2 + 3. \quad (76)$$

Proof. All fixed moments converge to the Gaussian moments by the cumulant formulas; normal moment determinacy identifies the limit. The logarithmic characteristic-function expansion is the cumulant series, uniformly on bounded t -sets when the parameters are sufficiently close to the corner. The displayed Edgeworth expression records the coefficient selected by the cumulant expansion. A uniform density remainder along a chosen two-parameter path additionally requires a path-uniform Fourier-tail estimate; the superpolynomial product bound supplies a natural route to such a refinement. $\square$

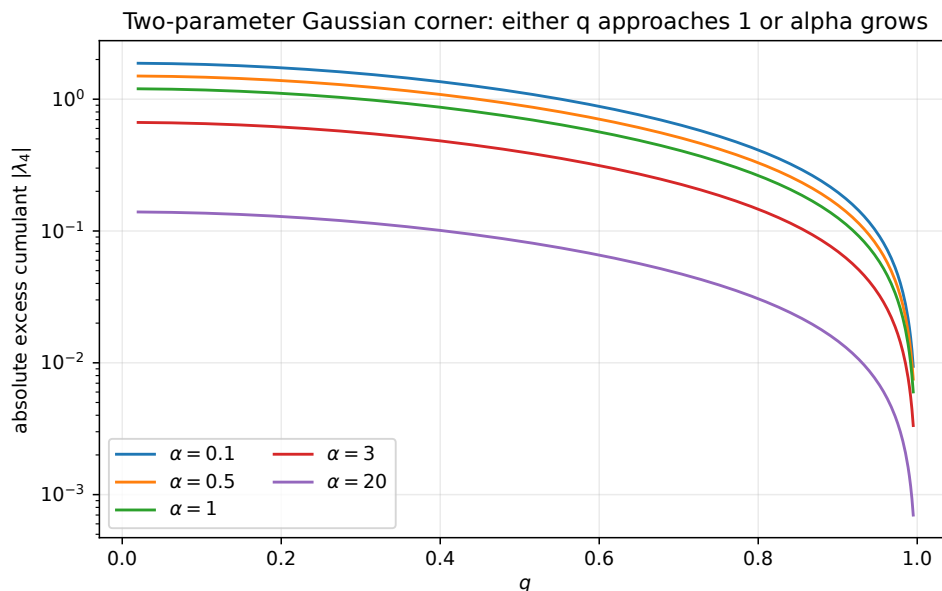


Figure 7: The exact magnitude of the excess fourth cumulant. It vanishes by either route: $q \uparrow 1$ or $\alpha \rightarrow \infty$.

9 Jacobi–Gegenbauer to Hermite orthogonal flow

Let $p_n^{(q,\alpha)}$ be the monic orthogonal polynomials of $\mu_{q,\alpha}$. Symmetry gives

$$p_{n+1}(x) = xp_n(x) - \beta_n(q, \alpha)p_{n-1}(x), \quad \beta_n > 0. \quad (77)$$

9.1 The two solvable edges

Theorem 9.1 (Jacobi edge and Hermite corner). *The family extends continuously to $q = 0$, where its measure is $\rho_\alpha(x)dx$, and*

$$\beta_n(0, \alpha) = \frac{n(n + 2\alpha - 2)}{4(n + \alpha - 1/2)(n + \alpha - 3/2)}. \quad (78)$$

This is the monic Gegenbauer/Jacobi recurrence, with the usual limiting interpretation at $\alpha = 1/2$, $n = 1$. At $\alpha = 1$, it is the monic Legendre recurrence.

After variance normalization, for every fixed n ,

$$\frac{\beta_n(q, \alpha)}{\sigma_{q,\alpha}^2} \longrightarrow n \quad (79)$$

whenever (q, α) approaches the Gaussian corner of Theorem 8.3. Hence the standardized monic polynomials converge coefficientwise to the probabilists' Hermite polynomials.

Proof. At $q = 0$, $X_{0,\alpha} = U_\alpha$, whose weight is the Gegenbauer weight with parameter $\lambda = \alpha - 1/2$. Converting the standard Gegenbauer recurrence [9] to monic form gives (78). At the Gaussian corner, every finite moment matrix converges to its Gaussian counterpart. Hankel determinant formulas for the monic recurrence coefficients are continuous because the limiting determinants are positive, and the Gaussian coefficients are n . $\square$

This is a two-parameter extension of the Legendre–Hermite bridge: q controls the number of effective geometric digits, while α controls the spherical/Jacobi shape of each digit.

9.2 Exact first coefficients

Put $v = \sigma_{q,\alpha}^2$. For any centered symmetric law with standardized cumulants λ_4, λ_6 , Gram–Schmidt gives

$$\beta_1 = v, \quad (80)$$

$$\frac{\beta_2}{v} = 2 + \lambda_4, \quad (81)$$

$$\frac{\beta_3}{v} = \frac{6 + 9\lambda_4 + \lambda_6 - \lambda_4^2}{2 + \lambda_4}. \quad (82)$$

Therefore

$$\frac{\beta_2(q, \alpha)}{v} = 2 - \frac{6}{2\alpha + 3} \frac{1 - q^2}{1 + q^2}. \quad (83)$$

Proposition 9.2 (Monotonicity of the second coefficient). *The normalized coefficient β_2/v is strictly increasing in both q and α , from its Jacobi value at $q = 0$ to the Hermite value 2 at $q = 1$.*

Proof. Differentiate the explicit expression (83). $\square$

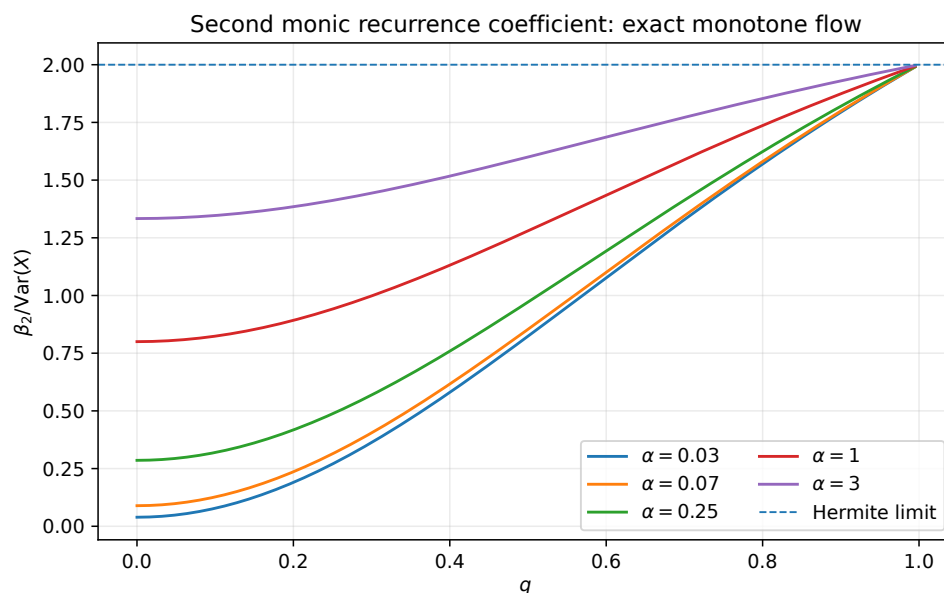


Figure 8: The exact monotone Jacobi-to-Hermite flow of the second normalized monic recurrence coefficient.

The third coefficient has a more surprising geometry. Set

$$A = 2\alpha + 3 > 3, \quad x = q^2 \in (0, 1). \quad (84)$$

Substitution of (69)–(70) into (82) and symbolic differentiation gives

$$\frac{d}{dx} \left(\frac{\beta_3}{v} \right) = \frac{36P_A(x)}{A(A+2)(1+x)^2(1+x+x^2)^2(Ax+A+3x-3)^2}, \quad (85)$$

where

$$\begin{aligned} P_A(x) = & A^3(x^6 + 4x^5 + 8x^4 + 10x^3 + 8x^2 + 4x + 1) \\ & + A^2(10x^6 + 44x^5 + 62x^4 + 20x^3 - 30x^2 - 28x - 6) \\ & + A(3x^6 + 32x^5 - 18x^4 - 74x^3 - 2x^2 + 48x + 11) \\ & - 6x^6 + 12x^3 - 6. \end{aligned} \quad (86)$$

The denominator in (85) is positive. At the endpoints,

$$P_A(0) = (A-1)(A-2)(A-3) > 0, \quad P_A(1) = 36A^2(A+2) > 0.$$

Thus any nonmonotonicity must be a genuine interior positive–negative–positive turning pattern.

Numerical observation 9.3 (Third-coefficient phase). The included minimization and resultant computation find a tangency at

$$\alpha_3^* = 0.066515108060995\dots, \quad q_* = 0.319296212569559\dots, \quad x_* = q_*^2 = 0.101950071361265\dots \quad (87)$$

For $0 < \alpha < \alpha_3^*$, P_A is negative on an interior interval; for sampled $\alpha > \alpha_3^*$, it is nonnegative on $[0, 1]$. The candidate $A_* = 2\alpha_3^* + 3$ is a real root of the degree-18 resultant factor listed in Section A.

Conjecture 9.4 (Sharp β_3 monotonicity threshold). *There is a unique threshold α_3^* given by (87) such that:*

- (i) *if $\alpha \geq \alpha_3^*$, then β_3/v is strictly increasing in $q \in (0, 1)$, except for the single horizontal tangency at equality;*
- (ii) *if $0 < \alpha < \alpha_3^*$, then β_3/v has one local maximum and one local minimum.*

A Sturm-sequence or cylindrical algebraic decomposition proof should be feasible because the problem has been reduced to positivity of the explicit bivariate polynomial (86) on a rectangle.

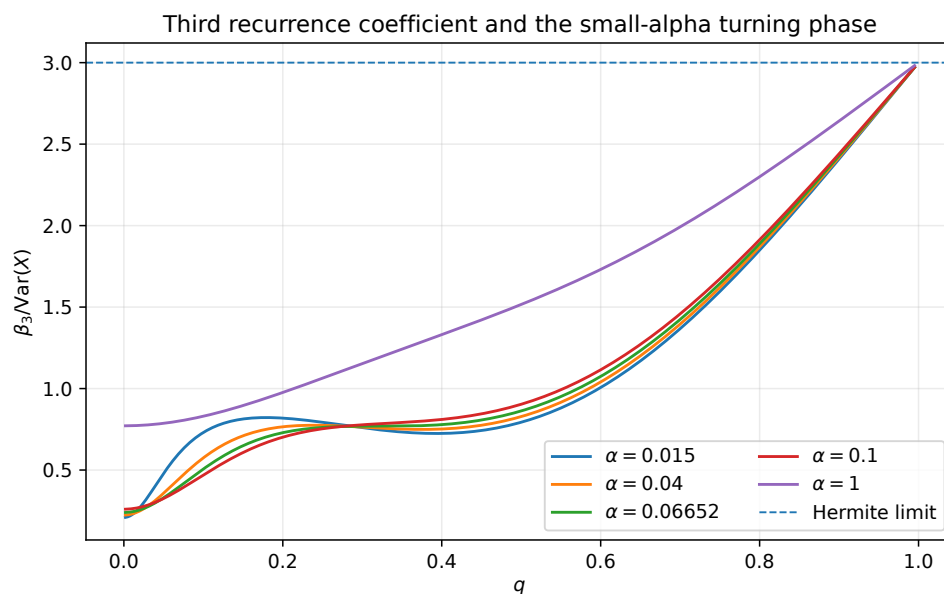


Figure 9: The third coefficient. Very small α produces a local maximum and minimum before the Hermite limit; the turning points merge at the observed threshold.

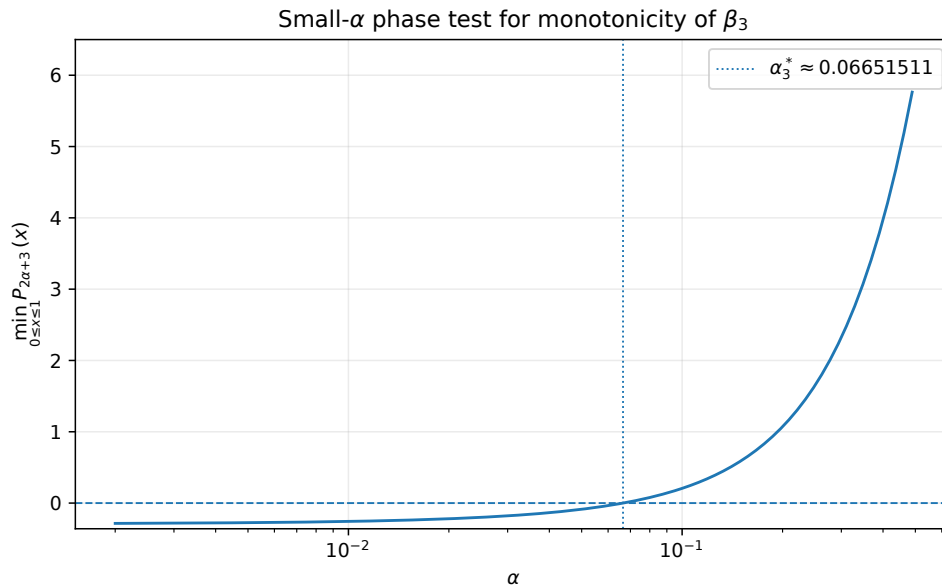


Figure 10: Minimum of the exact derivative-controlling polynomial $P_{2\alpha+3}$ over $0 \leq q^2 \leq 1$. The zero crossing identifies the candidate threshold α_3^* .

Conjecture 9.5 (Higher recurrence phase hierarchy). *For each $n \geq 3$, the normalized coefficient β_n/v has a finite collection of critical α -values at which its q -flow changes turning topology. These values are algebraic, obtained by eliminating q^2 from the numerator of $\partial_q(\beta_n/v)$ and its derivative. The sequence of largest thresholds tends to zero as $n \rightarrow \infty$.*

The last assertion is deliberately speculative; it is motivated by the idea that high-degree orthogonal polynomials resolve the increasingly fine Bernoulli-like structure near $\alpha = 0$.

10 Small- q and large- q expansions

The new family also provides controlled local expansions in the scale parameter. Write $\phi_\alpha(t) = \mathcal{J}_\nu(t)$.

Proposition 10.1 (Expansion at $q = 0$). *Uniformly for t in compact sets,*

$$\varphi_{X_{q,\alpha}}(t) = \phi_\alpha(t) - qt\phi'_\alpha(t) + q^2 \left[\frac{t^2}{2} \phi''_\alpha(t) - \frac{t^2}{2(2\alpha+1)} \phi_\alpha(t) \right] + O(q^3). \quad (88)$$

Thus the first perturbation differentiates the Jacobi characteristic function, while the second term contains the variance of the newly activated second digit.

Proof. In (13), expand the first factor $\phi_\alpha((1-q)t)$ to second order. The second factor has argument $(1-q)qt = qt + O(q^2)$ and equals $1 - q^2 t^2 / [2(2\alpha+1)] + O(q^3)$. All remaining factors contribute $1 + O(q^4)$. $\square$

At $q = 1 - \varepsilon$, the standardized cumulant factor satisfies

$$\frac{(1-q^2)^m}{1-q^{2m}} = \frac{2^{m-1}}{m} \varepsilon^{m-1} \left(1 + \frac{m-1}{2} \varepsilon + O(\varepsilon^2) \right). \quad (89)$$

Together with the Rayleigh recurrence, this generates the complete formal Edgeworth expansion to arbitrary order. The coefficients are rational in α , q -integer factors, and complete Bell polynomials; no Bessel zero computation is required.

At large α , for example,

$$\begin{aligned}\lambda_4(U_\alpha) &= -\frac{3}{\alpha} + O(\alpha^{-2}), \\ \lambda_6(U_\alpha) &= \frac{60}{\alpha^2} + O(\alpha^{-3}), \\ \lambda_8(U_\alpha) &= -\frac{3150}{\alpha^3} + O(\alpha^{-4}).\end{aligned}\tag{90}$$

These can be inserted into (68) to obtain a two-variable asymptotic expansion near the Gaussian corner.

11 Numerical experiments and reproducibility

The archive contains `experiments.py`, `requirements.txt`, all CSV outputs, and both PDF and PNG versions of every figure. The program uses:

- exact Rayleigh recurrences and Bell recurrences for cumulants and moments;
- 80-digit Hankel determinants for monic recurrence coefficients;
- corrected finite Bessel-zero sums as an independent check;
- fixed-seed Monte Carlo only for density visualization and a separate moment sanity check; and
- direct summation of the beta Laplace product to verify the periodic renormalization in [Theorem 7.3](#).

The final run used seed 20260830, 320,000 samples per density curve, and 600,000 samples per moment-validation parameter pair.

11.1 Validation results

The corrected sums of 240 Bessel zeros agree with the Rayleigh recurrence through R_5 to a worst relative error of 1.10×10^{-8} . Across the deliberately difficult parameter pairs listed below, empirical even moments through order eight agree with the exact Bell moments to a worst relative error below 5.3×10^{-3} .

α	q	order	exact moment	empirical moment	relative error
0.08	0.4	8	0.12213418	0.12200004	1.0983×10^{-3}
0.5	0.5	8	0.01426675	0.01424493	1.5294×10^{-3}
1.0	0.5	8	0.00408246	0.00410216	4.8254×10^{-3}
3.0	0.8	8	5.0084050×10^{-6}	5.0025652×10^{-6}	1.1660×10^{-3}

Table 2: Representative independent Monte Carlo checks. Exact full-precision values and actual empirical values are in `data/moment_monte_carlo_validation.csv`.

No theorem in the report depends on Monte Carlo. The numerical layer is used to discover the orthogonal phase, visualize the singular boundary, and catch algebraic or normalization errors.

11.2 Interpretation of the density figures

Figure 5 shows that the limit $\alpha \downarrow 0$ is not uniform in pointwise topology when $q < 1/2$: the densities remain smooth for every positive α but develop increasingly fine peaks separated by valleys. This makes the Wasserstein bound (66) a natural metric statement; uniform density bounds cannot hold.

At $q = 1/2$, Figure 6 displays the dimension ladder

$$S^0 \longrightarrow S^1 \longrightarrow S^2 \longrightarrow \text{high-dimensional spheres,}$$

projected through the same geometric random flight. The Rvachev law is not an isolated object in this picture: it is the $d = 3$ coordinate of a continuum of analytically continued spherical dimensions $d = 2\alpha + 1$.

12 Conjectures and frontier research program

12.1 Arithmetic and spectral questions

Conjecture 12.1 (Algebraic nonresonance). *Let $\alpha > 0$ be rational with $\alpha \neq 1$. For algebraic $q \in (0, 1)$, no equality*

$$q^d = \frac{j_{\alpha-1/2,k}}{j_{\alpha-1/2,\ell}}$$

occurs, except for explicitly classifiable half-integer Bessel orders with trigonometric reductions. Equivalently, the Fourier zeros are simple at algebraic scales outside a finite list of elementary cases.

Even proving irrationality of individual ratios of Bessel zeros is difficult, so this is a long-range target. A weaker first step is to establish nonresonance for almost every α at a fixed algebraic q , using analytic dependence of Bessel zeros on order.

Conjecture 12.2 (Smoothed spectral fluctuation). *A first Riesz mean of $N_{q,\alpha}(R)$ has an expansion consisting of a universal linear term, an α -dependent logarithmic term, and a bounded one-periodic function of $\log R/L$. Its Fourier coefficients are residues of $\Gamma(s)\mathcal{Z}_{q,\alpha}(s)$ at $s = 2\pi i\ell/L$.*

The unsmoothed sinc case contains binary digit sums, so smoothing is essential if one wants a bounded periodic remainder.

12.2 A matrix-valued Thue–Morse extension

Conjecture 12.3 (Jet cocycle and matrix automaticity). *For integer $m \geq 2$ and reciprocal-integer scales $q = 1/b$, repeated use of (43) can be organized as a finite-dimensional linear cocycle on a jet vector of $f_{q,m}$. The branch matrices are indexed by base- b digits. At $m = 1, b = 2$, their one-dimensional sign quotient is the Thue–Morse sequence; at $m > 1$, matrix products give a b -regular sequence controlling derivative signs and dyadic/radix rational values.*

A practical route is to choose the minimal jet closed under the two affine branches in (43), compute its semigroup, and ask whether it admits a triangular or representation-theoretic reduction.

12.3 Shape transition below $\alpha = 1$

Conjecture 12.4 (Unimodality and log-concavity curves). *There are critical curves $q_{\text{uni}}(\alpha)$ and $q_{\text{lc}}(\alpha)$ on $0 < \alpha < 1$ such that the cascade is, respectively, unimodal and log-concave above the curves. They satisfy*

$$q_{\text{lc}}(\alpha) \geq q_{\text{uni}}(\alpha), \quad \lim_{\alpha \uparrow 1} q_{\text{lc}}(\alpha) = 0,$$

and exhibit singular behavior as $\alpha \downarrow 0$ reflecting the Bernoulli-convolution phase at $q = 1/2$.

The Fourier product gives a direct numerical test through high-accuracy inversion, while total positivity or variation-diminishing convolution may supply proofs in part of the parameter plane.

12.4 Endpoint and inverse problems

The most immediate analytic problem is to prove [Theorem 7.6](#). The explicit periodic function (58) removes one major ambiguity: periodic coefficients are not fitted numerically but are convergent analytic sums. A complete solution should proceed through a uniform saddle theorem for geometrically self-similar Laplace products, followed by transseries reversion for $Q_{q,\alpha}$.

Further questions include:

- (a) determining how the fractional order α enters the first logarithmic and log log corrections to (61);
- (b) proving differentiability of the periodic profile with respect to both q and α , including limits at integer α ;
- (c) finding Stokes phenomena in the complex endpoint plane; and
- (d) matching the $\alpha \downarrow 0$ endpoint expansion to large deviations of Bernoulli convolutions.

12.5 Orthogonal and interpolation questions

The exact moment recurrence (30) makes the Hankel determinants amenable to symbolic computation. Promising targets are:

- (a) a Sturm/CAD proof of [Theorem 9.4](#);
- (b) exact resultant phase diagrams for β_4, β_5 , followed by asymptotics in n ;
- (c) Christoffel-function estimates uniform in the Jacobi–Hermite corner;
- (d) geometric-comb interpolation in the basis $p_n^{(q,\alpha)}$, comparing Lebesgue constants with the existing Legendre and native-up constructions; and
- (e) a double-scaling universality regime in which $n(1 - q)$ and n/α remain finite, potentially producing new transition kernels between Jacobi and Hermite asymptotics.

12.6 Vector random-flight questions

The vector law (6) opens a separate geometric frontier. At $q < 1/2$, its support has a flat inner boundary as well as a flat outer boundary. One can ask for coupled asymptotic expansions at both radii, including the critical coalescence at $q = 1/2$. Spherical harmonics should diagonalize angular modes of the refinement operator, with Gegenbauer polynomials governing coordinate projections. This may produce a natural multidimensional analogue of Rvachev atomic functions on balls and annuli.

12.7 Formalization plan

A staged Lean formalization can avoid measure-theoretic obstacles at first:

1. formalize the rational Rayleigh recurrence and generate (23)–(27);
2. formalize the algebraic cumulant scaling and Bell recurrence;
3. prove the moment fixed-point recurrence (30);
4. verify the exact β_2 and β_3 formulas by polynomial normalization;
5. use a certified real-algebraic package for the $P_A(x)$ positivity problem; and only then

6. add probability measures, infinite convolutions, and distributional derivatives for the analytic theorems.

The finite algebraic core is already well suited to exact computation and would provide reusable infrastructure for the repository’s existing Bernoulli–Bell and orthogonal-polynomial developments.

13 Conclusion

Changing the scale parameter alone preserves the uniform digit and therefore preserves the sinc atom. Changing the digit to the Jacobi family exposes a larger structure:

$$\begin{array}{c} \text{spherical geometry} \longleftrightarrow \text{Bessel zeros} \longleftrightarrow \text{Rayleigh cumulants} \\ \longleftrightarrow \text{Jacobi orthogonality} \longleftrightarrow \text{fractional endpoint order} \end{array}$$

The Rvachev law sits at the particularly symmetric point $d = 3, \alpha = 1$, while the Bernoulli convolution is the boundary $d = 1, \alpha = 0$, and Hermite behavior appears at high dimension or high digit density. The same geometric series produces a q -integer spectral factor, a dense resonance set, and log-periodic endpoint corrections.

The most robust new results are the exact formulas: the Bessel product, Rayleigh cumulants, spectral-zeta factorization, higher-order refinement, and fractional pantograph. They are independent of numerical evidence and reduce at $\alpha = 1$ to known repository formulas. The numerics then reveal a less expected phenomenon: orthogonal recurrence coefficients remember the near-Bernoulli microstructure, producing a genuine turning phase long before the density ceases to be smooth. That phase, the complete Lambert expansion, and the matrix Thue–Morse cocycle form a coherent next research program.

A The resultant polynomial for the β_3 phase

Eliminating x from $P_A(x) = 0$ and $\partial_x P_A(x) = 0$ produces several extraneous or boundary factors and the following degree-18 factor:

$$\begin{aligned} Q(A) = & 27A^{18} + 324A^{17} - 5670A^{16} - 58320A^{15} + 2599461A^{14} + 23888916A^{13} \\ & - 213002432A^{12} - 1553944800A^{11} + 10096215141A^{10} + 40741963244A^9 \\ & - 197857358766A^8 - 118077559920A^7 + 579333364587A^6 + 348162442236A^5 \\ & - 247263068556A^4 - 54019599840A^3 + 57183051600A^2 - 12550510656A + 918330048. \end{aligned}$$

It has one real root greater than three,

$$A_* = 3.133030216121990\dots, \quad \alpha_* = (A_* - 3)/2.$$

The other real roots are negative. This algebraic isolation strongly supports [Theorem 9.4](#); a complete proof must additionally certify the sign of P_A over the two parameter cells separated by the tangency.

B A direct formula for the periodic profile

For reference, the proof of [Theorem 7.3](#) gives a numerically stable algorithm. Given $\theta \in [0, 1)$, sum the negative-index part of (58) until $e^{L(m+\theta)}$ is below tolerance; sum the corrected positive-index part until its $O(e^{-Lm})$ tail is below tolerance. No oscillatory integration or Mellin inversion is needed. Differentiation with respect to θ is termwise, so Fourier coefficients can be obtained by an ordinary periodic quadrature.

The source program implements a complementary check: it evaluates $\mathcal{K}(e^{L(N+\theta)})$ directly for several N and subtracts the quadratic and linear terms. The collapse in [Figure 4](#) confirms both the coefficients and the phase convention.

C Numerical file inventory

File	Purpose
experiments.py	Reproducible symbolic, high-precision, spectral, and Monte Carlo computations.
data/exact_digit_cumulants.csv	Symbolic cumulants through order ten.
data/standardized_cumulants.csv	Grid of $\lambda_4, \lambda_6, \lambda_8, \lambda_{10}$.
data/orthogonal_recurrence_coefficients.csv	High-precision $\beta_1, \dots, \beta_5$ and variance-normalized values.
data/beta3_monotonicity_phase.csv	Minimum of P_A and its argmin over a dense α grid.
data/beta3_threshold_polynomial.txt	The resultant factor and numerical tangency.
data/rayleigh_sum_validation.csv	Independent Bessel-zero checks of the Rayleigh recurrence.
data/resonance_cloud.csv	Finite samples of the resonance set.
data/resonance_density.csv	Maximum-gap evidence for density.
data/bernoulli_boundary_W1_ratio.csv	Gamma-ratio bound and convergence to $2 \log 2$.
data/endpoint_laplace_periodic_profile.csv	Direct periodic-profile convergence data.
data/moment_monte_carlo_validation.csv	Independent fixed-seed moment checks.
figures/*.pdf, figures/*.png	Vector and raster versions of all report figures.

D Repository comparison record

The nearest directly inspected repository sources were, at the pinned snapshot:

- semi-formalized-research-frontiers/drafts/MANIFEST.md;
- .../exponents-and-q-series/Exponents_and_q_Series_Frontiers/ Exponents_and_q_Series_Frontiers.tex;
- .../Cyclotomic_q_Fabius_Rvachev_Frontier/ cyclotomic_q_fabius_frontier.tex;
- .../Fabius_Flat_Parameter_Response_Dynamics/;
- .../Atomic_Functions_Beyond_Dyadic_Frontiers/; and
- the canonical representation, inverse-endpoint, and orthogonal summaries referenced by the manifest.

The decisive distinction is that those reports deform the scale, exponent sequence, representation, or response of the uniform-digit law, whereas the present report deforms the digit law itself through the Jacobi parameter.

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